producers attempt to feed dry, moisture-consistent materials to the plant whenever possible. A solid sloped or even a paved draining surface under the aggregate stockpile area greatly helps achieve this objective.

7.2.3 Virgin Aggregate and RAP Cold-Feed Bins

Feed bins for virgin aggregate (see Figure 7.6) and RAP (see Figure 7.7) are similar for both batch and drum-mix plants.

While vibratory pan feeders and plate-style feeders were used in the past, virtually all plants, both batch and drum/continuous, now use conveyors with variable-speed motors located below feed bins to regulate the flow of materials to the drying drum.

Variable-speed belt feeds are consistent and repeatable, and therefore easy to calibrate. The ton-per-hour output can be determined at different belt speeds, and then the required speed setting can be calculated for a given recipe and given production rate. The feed-bin calibration process is the act of measuring and recording output at different speed ranges for a specific gate opening. This process is usually computerized on today's modern plants. Gate openings should be proportional to the amount of aggregate being added to the plant to ensure uniform loading of the belt.

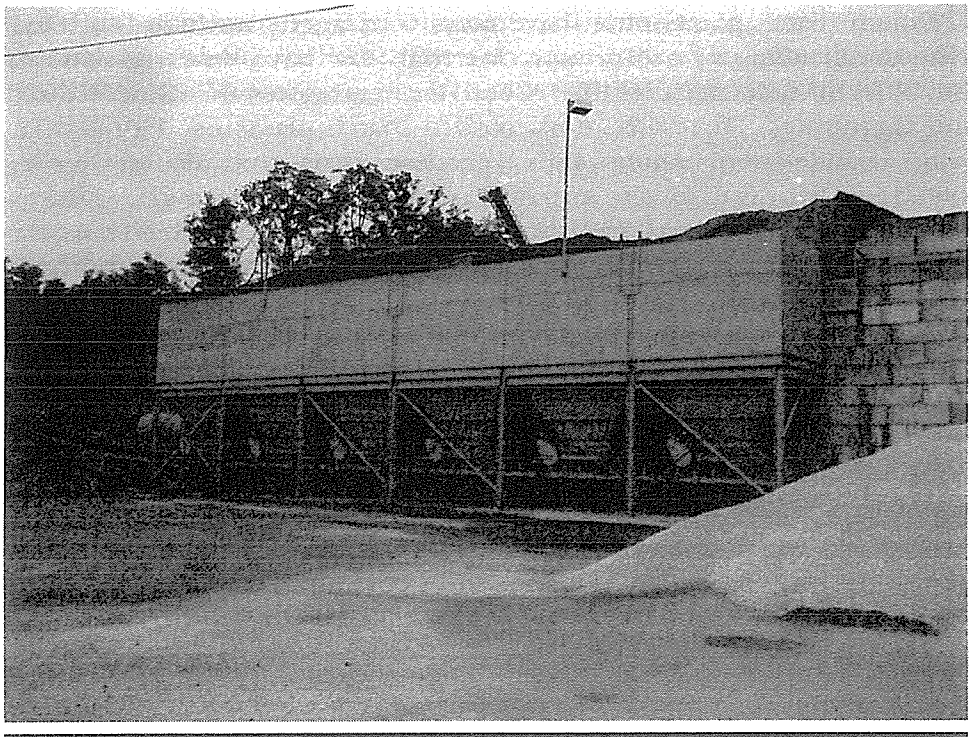


Figure 7.6 Cold-feed Bins with Belt Feeders

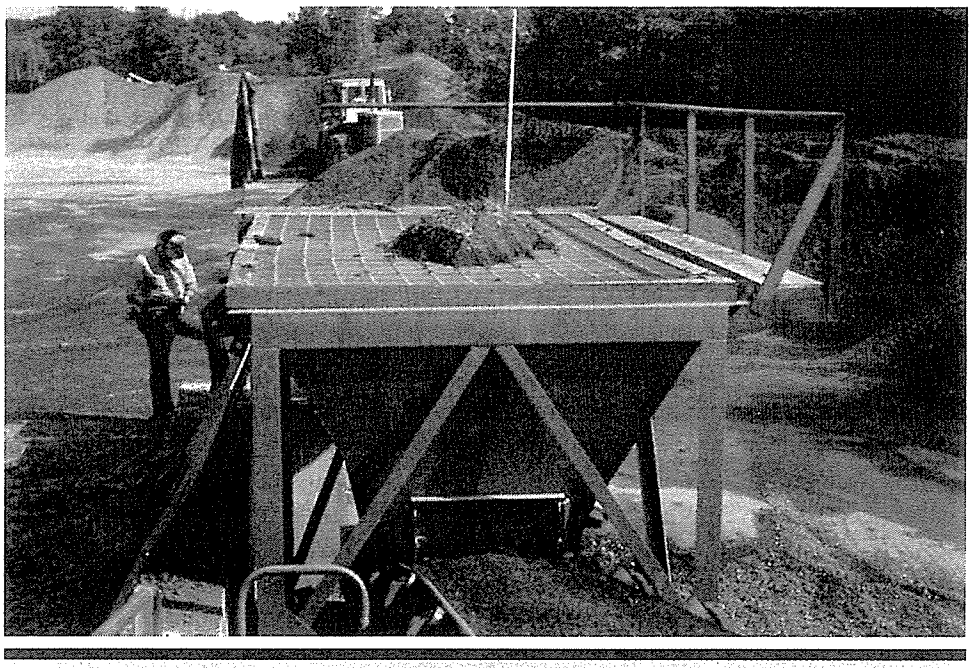


Figure 7.7 Bin for Reclaimed Asphalt Pavement

With a drum-mix plant, setting the different feeder speeds actually establishes the gradation blend in the final mix.

Modern batch plants often have more feed aggregates than hot bins, with the blending of two or more materials that have similar gradations but different characteristics (like natural fine aggregates and manufactured fine aggregates). Therefore, most modern batch plants use belt feeders with variable-speed motors and calibration charts and also set feeder speeds for their feed bins to keep final gradation accurate, much as drum-mix plants do. When final gradation is controlled at the cold-feed bins, the screening unit in the plant is usually bypassed to increase production and reduce screen wear.

It is important that the calibrated output of these belt feeders be checked periodically for maximum mix accuracy. Changes to the speed/output relationship can occur because of buildup inside the feed bins or fatigue to the motor/drive unit. Some regulatory agencies require calibration checks at certain intervals or before the start of large projects. Calibration accuracy should be checked on a regular basis, regardless of requirements.

7.2.4 Aggregate and RAP Transfer Conveyors

Both batch and drum-mix facilities use the same style of aggregate conveyor to transfer material between the cold-feed bins and the plant (see Figure 7.8). These are belt-type conveyors with fixed speeds. They can

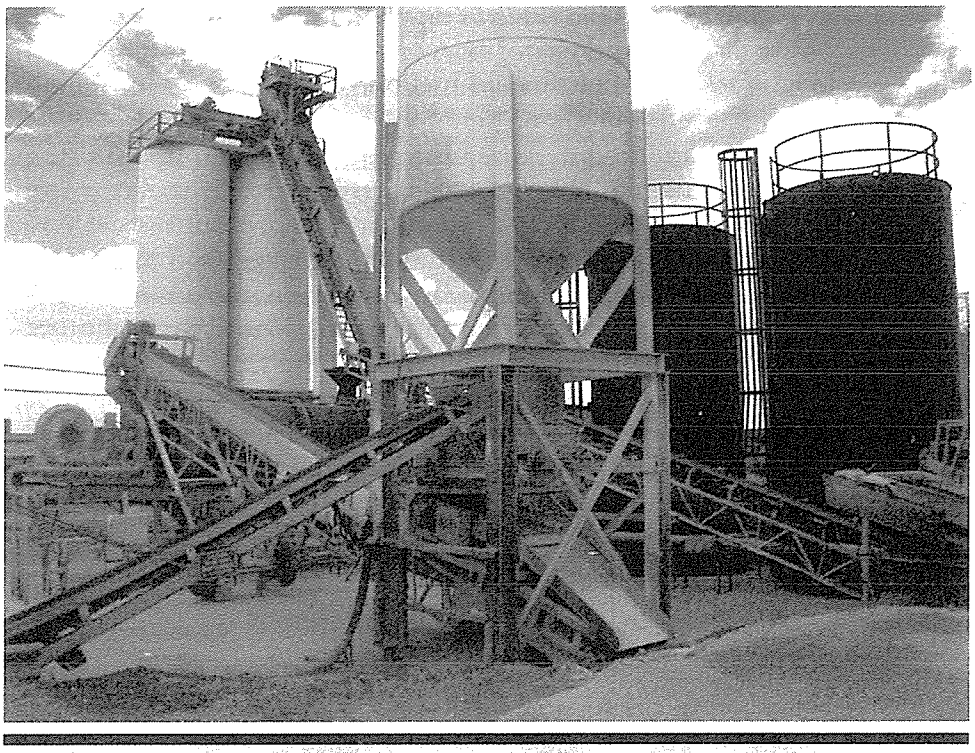


Figure 7.8 Aggregate Conveyors on a Drum Plant

be fitted with scrapers or guards to minimize spillage or with wind guards to reduce the effect of the wind on belt scales.

7.2.5 Aggregate Dryer

An aggregate dryer is needed for both batch and drum-mixer plants.

With some styles of drum-mixers, the dryer is very long because it is also used as a mixing drum to mix the liquid asphalt binder with the aggregate to produce the final mix. Some drum-mixers are two piece units—the dryer and the mixing device are separate units.

It is the responsibility of the dryer to produce mix with a consistent temperature. Wide variations in temperature in the final mix affect placement technique and in-place density. A temperature swing greater than 10°–20°F (6°–11°C) typically affects field performance. Aggregate moisture that varies widely in the stockpiles can make meeting this directive difficult for the plant operator. This is one reason why managing the stockpile area for moisture control is so thoroughly discussed.

Aggregate dryers and drum-mixers come in both counterflow configurations (aggregate and hot gases traveling in opposing directions) and parallel-flow configurations (aggregate and hot gases traveling in the same direction). Batch plants have counterflow dryers. The newer drum-mixers discussed in this chapter also utilize counterflow dryers.

Dryers have a burner and fan fitted at one end to create a hot gas stream that dries and heats the aggregate. Burner and dust control equipment allow dryers with fossil-fuel burners to function very effectively from both a particulate- and gaseous-emission standpoint.

Dryers are inclined and rotate. They function by picking up and dropping aggregate material into the hot gas stream. Inside the dryer are steel flights, or lifters. These flights pick up the aggregate material and drop it through the hot gas stream as the dryer rotates (see Figure 7.9). To prevent incomplete combustion and hydrocarbon contamination, the aggregate material never passes through the burner flame. Special burner flighting is installed in the dryer for the length of the flame. These special flights pick up the aggregate, trap it against the outside wall of the dryer, and carry the material over top of the flame. Gravity causes the material to advance slightly with each rotation because the dryer is installed on a slight incline. Retention time typically is three to five minutes but can be longer, depending on the dryer's size, length, and flight configuration. Retention time can be lengthened for extremely absorptive, wet, or



Figure 7.9 Flights Lift and Drop Material Through the Hot Gases of the Dryer

otherwise difficult-to-dry material by installing dams in the dryer between the rows of flights or by changing the angle of incline of the dryer if the unit is adjustable.

Flights can be configured for different material and local conditions. Small aggregate particles heat quickly and easily, although they are typically wetter and require a higher burner position and exhaust fan damper position to remove the additional water from the rock. Coarse mixes with large quantities of stone have more mass for each particle and often require a more aggressive flight pattern for maximum heat transfer into the stone. The equipment manufacturer can provide advice for adjusting flight configurations.

Flights also wear as they dig through the aggregate material on the bottom of the dryer. As their leading edges wear down, drying efficiency is reduced. If moisture is retained in the mix, quality and placement will be compromised and the long-term serviceable life of the pavement may be affected.

Moisture retention in the mix can be monitored simply by weighing produced mix, drying it, and then weighing it again. Any weight loss can be attributed to moisture in the aggregate material. Reducing the feed rate will usually solve the problem.

Elevated exit gas temperature is another indication that the flights are either worn or that the configuration is wrong for local conditions. Flights can easily be replaced if they are worn. The pattern of the flight can also be changed for more drying efficiency.

Another indication that the dryer is functioning inefficiently for project conditions is a significant drop in material temperature after production. If the aggregate in a batch plant—or the mix in a drum-mixer plant—loses temperature significantly between the time it leaves the dryer and is loaded into the truck during normal production (more than 25°–30°F/14°–17°C), the aggregate is either not completely dry or not heated all the way through. Mixes with a high percentage of large aggregate often require elevated dryer exit temperatures to compensate for the longer time it takes for the colder center of the stone to heat-stabilize, but a temperature drop might also indicate that the stone is not completely dry. When significant temperature drops are encountered, moisture retention tests are needed. If no moisture is present, the uncoated stone can be overheated to compensate for the eventual temperature drop.

Drying effectiveness should be monitored during production and over the life of the equipment. The flights in the dryer are subject to wear because they constantly dig through and handle aggregate materials. These flights will wear and must be maintained. Their maintenance is part of the inspection/monitoring process for plant operating efficiency. Recording the surface temperature taken with an infrared temperature gun along the length of the dryer shell is an excellent way to monitor the flighting arrangement and drying efficiency. By plotting the temperature readings at 2-foot (0.7-meter) intervals, hot and cold spots along the length of the

dryer will indicate where the problems are located inside the dryer. The temperature plot should increase uniformly along the length of the dryer shell to confirm that the dryer is working properly.

7.2.6 Dust Collection Equipment

The gas velocity in an aggregate dryer is quite significant and will remove fine aggregate fractions from the dried aggregate mixture. Dust collectors are installed on hot mix plants primarily to collect the dust for environmental reasons, but captured dust is often reused, depending on the type of equipment installed on the plant.

There are two stages of dust collection on plant systems—*primary collectors* and *secondary collectors*. Primary collectors gather the large dust/fine aggregate particles. Secondary collectors gather the smaller particles.

Most primary collectors are either the knockout-box type or the cyclone type. Figures 7.10 and 7.11 show these types of collectors installed on plant systems. The fines collected in primary collectors are returned 100 percent to the plant process, without any contemplation of partial return. Most people consider this dust to be sand-size particles, but in reality they also have a fairly high percentage of material passing the 200-mesh or 75-micron sieve. The same style of primary collectors are found on both batch and drum-mix facilities.

Secondary dust collectors fall into two categories—wet washers (or wet scrubbers) and baghouses (or fabric filter collectors). This equipment is positioned immediately upstream of the exhaust fan and is connected to the fan with ductwork. Figures 7.11 and 7.12 show these types of systems installed on hot mix plants.

Wet washers clean the dust from the gas stream by injecting additional water into the system. This makes the dust particles heavier, allowing them to fall out of the gas stream. Collected dust is very wet—basically a wet sludge—and so cannot be returned for use in the mix process. This effluent is pumped to a settling pond where the dust settles to the bottom. The particle-free water is then recycled from the surface of the pond and reused for dust collection.

A baghouse cleans with a dry process, allowing the gas to pass through a fabric filter supported by a wire cage. Dust is collected on the outside of the fabric filter while the dryer is running, and cleaning air periodically passes through the bag from the opposite direction, causing the collected dust to fall into the hopper of the baghouse. This collected dust can either be returned to the mix-production process or removed from the system. Figure 7.13 shows the internal components of a baghouse.

When dust is returned to the plant process, certain operational practices are important in order to keep the dust return consistent.

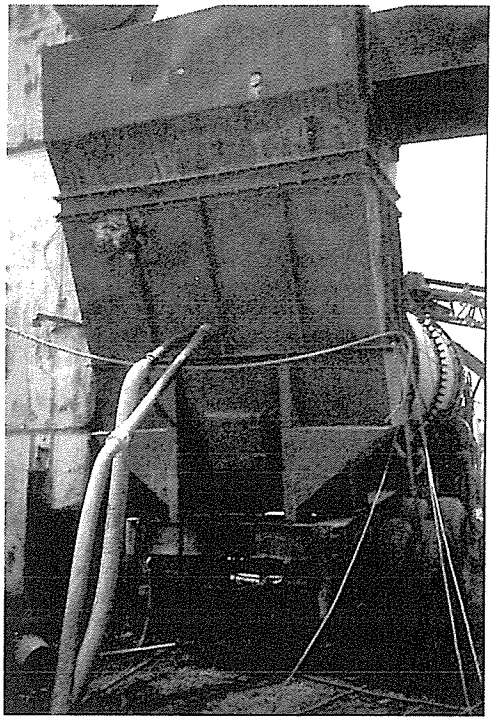


Figure 7.10 Knockout Box on Drum-Mixer Plant

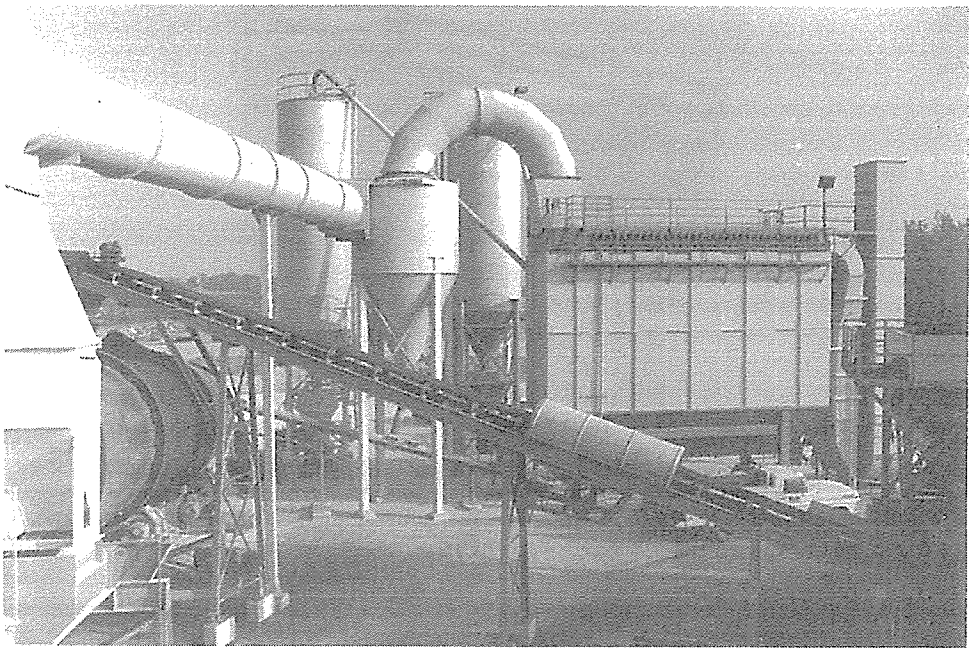


Figure 7.11 Cyclone and Baghouse on Drum-Mix Plant with Dust Return and Mineral Filler Silos

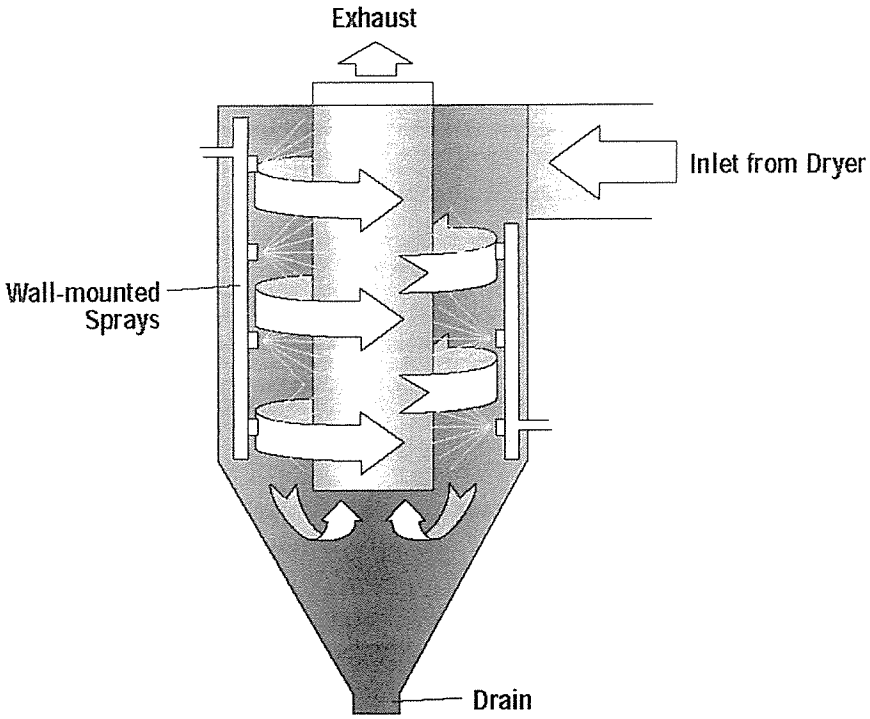


Figure 7.12 Scrubber on Drum-Mix Plant

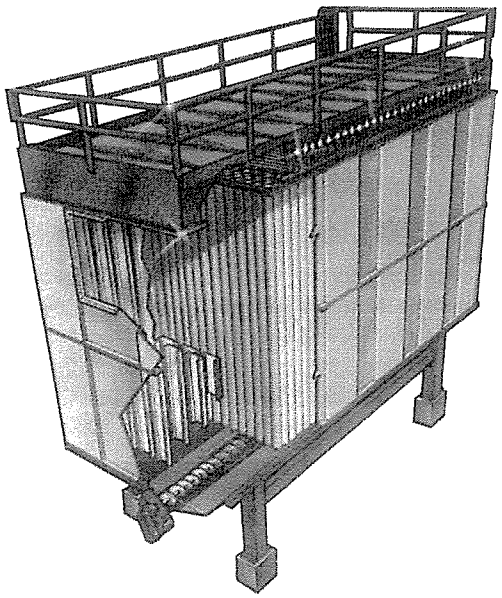


Figure 7.13 The Internal Components of a Baghouse

Section 7.2.7, "Dust-Return Equipment," discusses several best operating practices to ensure that the dust return is consistent with the dust removed and that voids in the mix are not negatively affected.

7.2.7 Dust-Return Equipment

Most plants are fitted with fabric filter collectors, allowing operating personnel to return the captured dust to the mix-production process. As mentioned above, fines collected from primary-type collectors are returned 100 percent to the process without any contemplation of partial return (see Figure 7.14).

Dust from a baghouse secondary collector, however, can be returned entirely or in part, depending on the requirements of the job-mix formula. Fines return is accomplished mechanically with screw conveyors or pneumatically with compressed air or blower systems.

The general practice is to design mixes so all the fines collected in the baghouse are reused in the mix or so all the fines are wasted/removed from the process. This keeps the equipment options and operating principles simple.

When 100 percent of the dust collected in the baghouse is to be reused, the following operational best practices are important:

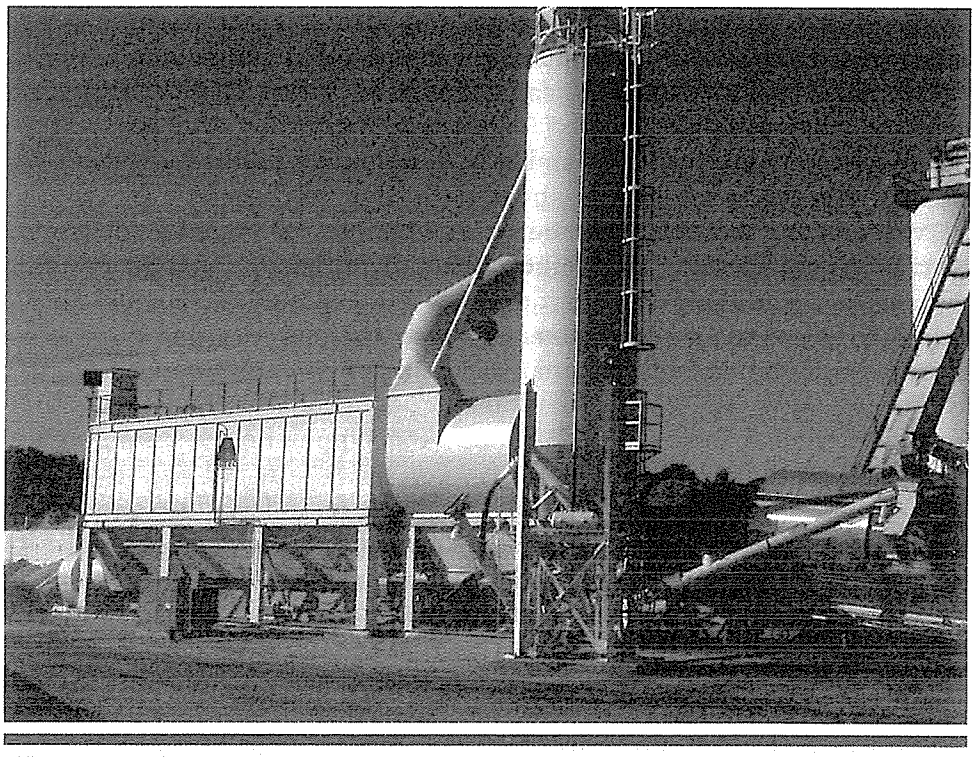


Figure 7.14 Fines-Return System on Drum-Mix Plant

- The baghouse cleaning and fines return must stop when the dryer stops. This keeps large quantities of dust from being transferred to the mixing area of the drum or deposited in the fines bin on the batch plant when the aggregate feed stops.
- Large and rapid increases in production rate should be avoided. It takes time for the dust to be collected on the bags, cleaned off the bags, and then returned to the system. Large jumps in production rate can be immediately accomplished by resetting the cold feed bins, but it takes time for the dust corresponding to the production-rate increase to be collected on the bags and returned to the mix. The mix produced during that time will be short of –200 mesh (75 micron) material and high in voids. Large, rapid decreases in production have the opposite effect. Most modern control systems ramp production up and down in small intervals because of this.
- Fines-return equipment inspection should be done regularly. When the fines-return equipment wears, it may only be able to return a certain quantity of fines. Test results are characterized by a shortage of fines and an increase in voids above a certain rate on the plant system.

When partial dust return is required, several equipment options exist for both batch plants and drum-mix plants. Most plant owners purchase a storage silo for transferring the collected fines for reuse or for storage for later use. If the silo fills to capacity, accumulated fines are usually trucked off and wasted or used for other purposes, such as in the manufacture of slurry seals. Figures 7.14 and 7.15 show this type of equipment.

Baghouse dust stored in silos can either be reintroduced into the process volumetrically with a calibrated feeder valve (with output charted to the speed of the feeder valve, as similarly done in the calibration process of cold feed bins), or it can be weighed back into the process.

With a batch plant, dust can be returned with a calibrated feeder to the hot aggregate elevator, be weighed back through the aggregate weigh hopper as an additional hot bin draw, or be weighed in a separate scale and then deposited into the mixer. All these options are common.

In a drum-mix plant, material can similarly be returned through a calibrated valve, or a negative-weighing pod can be used to continuously weigh dust back. Figure 7.16 shows this type of negative weigh pod system.

Local specifications often determine which method of dust reentry is desirable or required. The important point is that many options exist for returning captured dust to the mix production process. Mix requirements and local specifications dictate the required equipment options.



Figure 7.15 Fines-Return System with Storage Silo

7.2.8 Mineral Filler Additive Storage and Feed Silos

When additional mineral filler additive is required, the same types of silos and conveying systems for handling excess baghouse fines, as previously

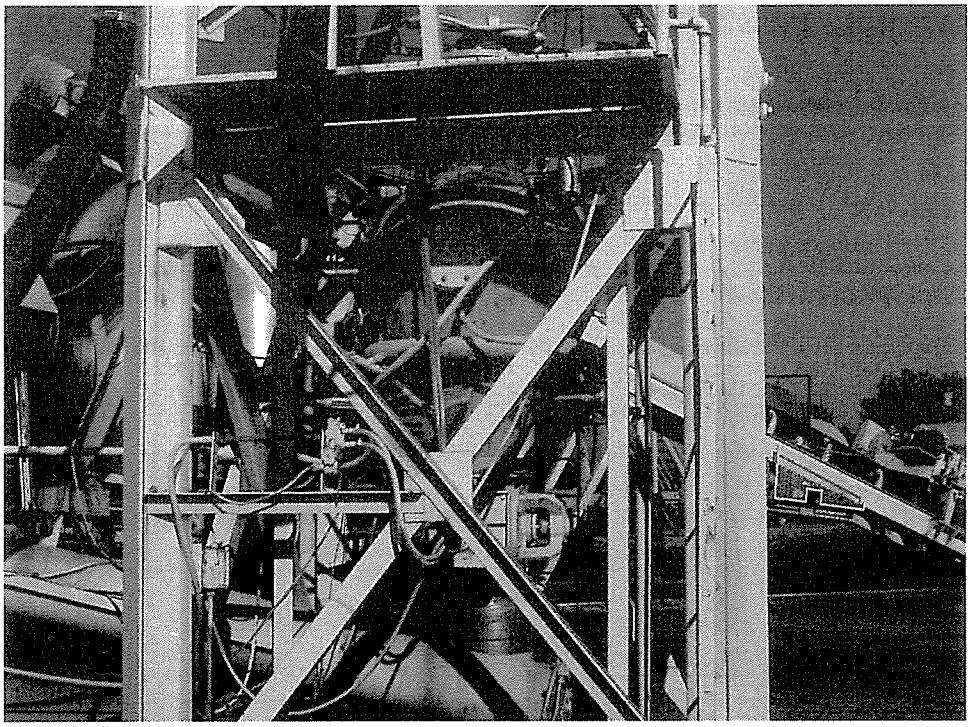


Figure 7.16 Continuous Weigh Pod for Baghouse Dust or Filler on Drum Plant

discussed, are typically used. Filler additives can be added either volumetrically or by weight with the same type of equipment. Plant automation handles the addition of this material as part of the mix formula.

7.2.9 Lime Additive Bulk Storage and Feed Silos

Some specifying agencies require certain aggregates to be treated with lime for antistripping control.

Bulk lime is usually delivered to the plant site in tankers, transferred pneumatically with a blower into a storage silo, and then introduced in ways similar to that of a bulk mineral filler additive. Depending on local requirements and preferences, lime is either introduced into the mixing section of the plant or mixed with the aggregate before the drying process. When the lime is introduced into the mixing area of the plant, the equipment shown in Figures 7.14 and 7.15 can be used. When lime is mixed with aggregate before the drying process, equipment such as that shown in Figure 7.8 is often used. Lime can also be premixed with the aggregate before shipment to the plant site.

The control routine is similar to that previously described for bulk mineral filler additives.

7.2.10 Fiber Additive Equipment

Fiber additives are specified for some hot mix formulas to help ensure that a uniform film thickness is maintained between the aggregate particles and to reduce the possibility of draindown of the binder in the mix. They are produced from both mineral fiber and cellulose fiber materials.

Fiber additives are popular with mixes that contain a high percentage of large-size or same-size aggregate and a low percentage of fine aggregates to fill the voids between these large aggregate particles—like Open-Graded Friction Course (OGFC) and stone matrix asphalt (SMA) mixes.

Fibers are typically added pneumatically with a machine that shreds and blows the fibers into the mixing area of the plant. This machine uses a negative-weighing principle where the bundle of fibers deposited in the machine is continuously weighed, and the declining weight is calculated and displayed on the control system as it is separated and blown into the plant.

In a drum-mixer application, this machine runs continuously and there is a continuous weight display of the fiber additives. The flow rate of the machine is typically tied into the plant master controls and the aggregate belt scales.

In a batch-plant application, the machine starts and stops based on the target weight of fibers for each batch. With a batch plant, fibers can also be added by hand in bag form by dropping a pre-packaged bag of fibers down a chute directly into the pugmill with each batch.

7.2.11 Finished Product Storage Silos and Load-Out Equipment

Finished product storage silos allow the plant to run continuously whether or not trucks are present. Silos are required for drum-mix plants, because the mix flows continuously from the mixing area when the plant is running and must be elevated for dispensing into trucks. Silos on batch plants allow the plant to run continuously, thereby increasing daily output.

Silos can be either the surge or the storage type. Surge silos are typically not insulated and are often open-topped bins. This type of storage unit is lighter and can be made portable so that it works well with portable plants. Storage silos are fully enclosed and heavily insulated. They generally have heated cones, and some have sidewall heat. Top and bottom seals are common on storage silos. Storage silos allow longer storage times for mixes, without significant heat loss or oxidation to the mix. Storage silos are massive units requiring large foundations, making them semi-permanent installations. Regulatory agencies sometimes rate storage silos for storage periods or limit maximum storage times. Allowable storage times are sometimes based on the absorption characteristics of the aggregate, rather than on the features of the plant equipment.

Portable silos commonly have load cells built into them, allowing trucks to be loaded and ticketed on a negative-weigh basis. Figure 7.17 shows such a silo.

Silos at stationary plants are usually mounted over truck scales, allowing the tare weight, gross weight, and net weight to be captured with each load. Multiple silos and multiple scales are not uncommon. Figure 7.18 shows such a system.

The use of a batcher or a gob hopper is a good practice and is generally required by specification for silos of any type. Batchers collect mix in small quantities at the top of the silo and drop the mix in mass into the silo. This action minimizes the segregation that can occur when silos are fed continuously. Batchers and gob hoppers need to be adjusted so that the maximum size batch is dispensed into the silo each time. These devices typically control segregation on most traditional mix designs, but some mixes are especially prone to segregation, especially if they are "gap-graded" or have especially large rock in them. Segregation can also occur in the batcher/gob hopper itself if the device is not fed directly in the center, fed obliquely at an angle, or positioned significantly below the feeding device.

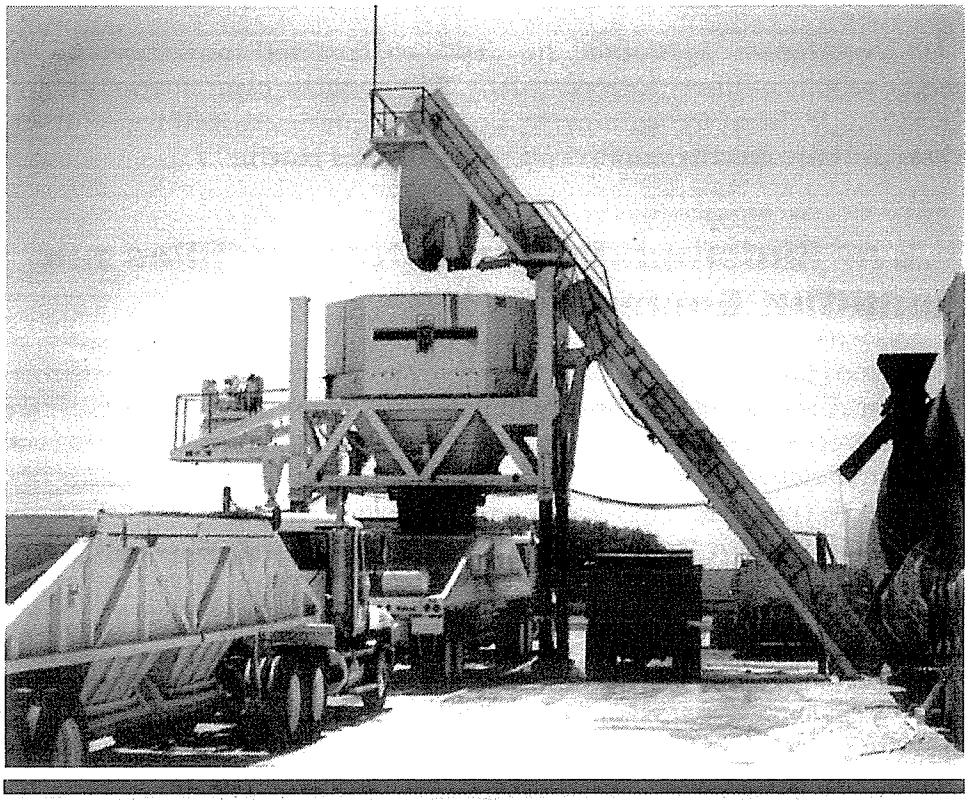


Figure 7.17 Self-Erecting Silo with Weigh Scale in Silo

Heating elements (on the silo walls) hotter than the mix temperature may damage the mix by overheating it at the point of contact, and may also cause it to build up on the silo walls. This buildup, in turn, ultimately acts as an insulator, which reduces the long-term storage capability of the silo. Heat should be set at—or at no more than 20°–30°F (11°–17°C) higher than—the mix temperature.

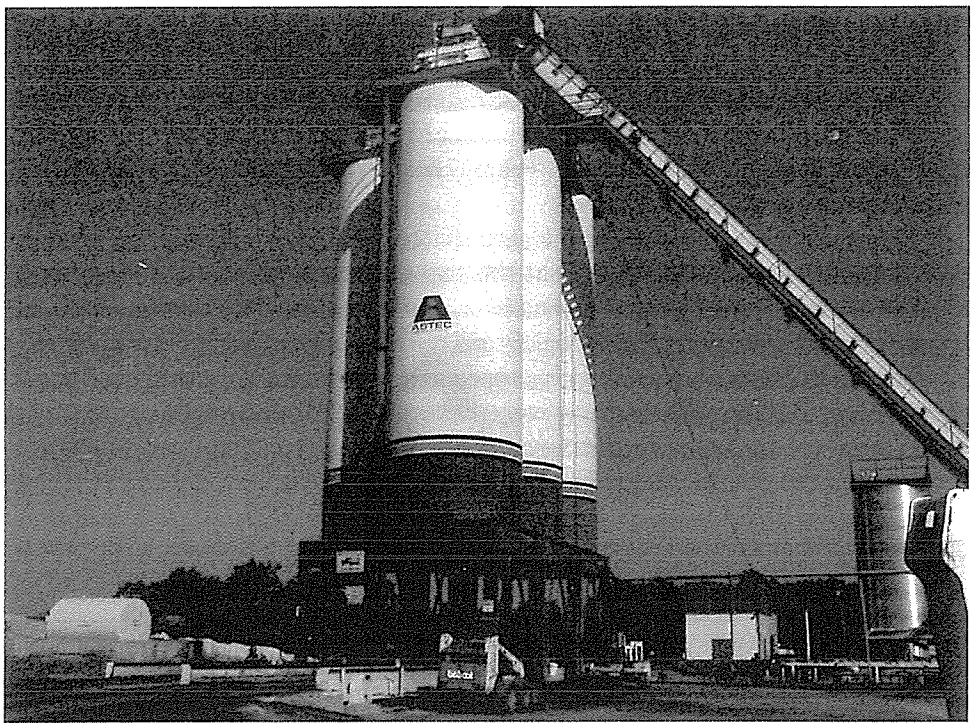


Figure 7.18 Stationary Silos over Truck Scale

7.3 The Batch Plant Production Process

7.3.1 Aggregate Gradation Control with a Batch Plant

Aggregate gradation is controlled in two places in a modern batch plant—in the cold feed and in the screening unit/hot bins (see Figure 7.19). Changes in the past few years have complicated the gradation control process in a batch plant.

Batch plants were originally designed to accept a crushed aggregate product and a sand. The screens at the top of the batch tower were used to separate the aggregate into fractions so that different mixes could be constructed, some with larger rocks and a larger percentage of stone for base or binder lifts and some with more sand and smaller stones for wearing surfacees. When batch plants were first designed, purchasing presized aggregates for hot mix production was not an option. The batch tower's first job was to size the aggregates.

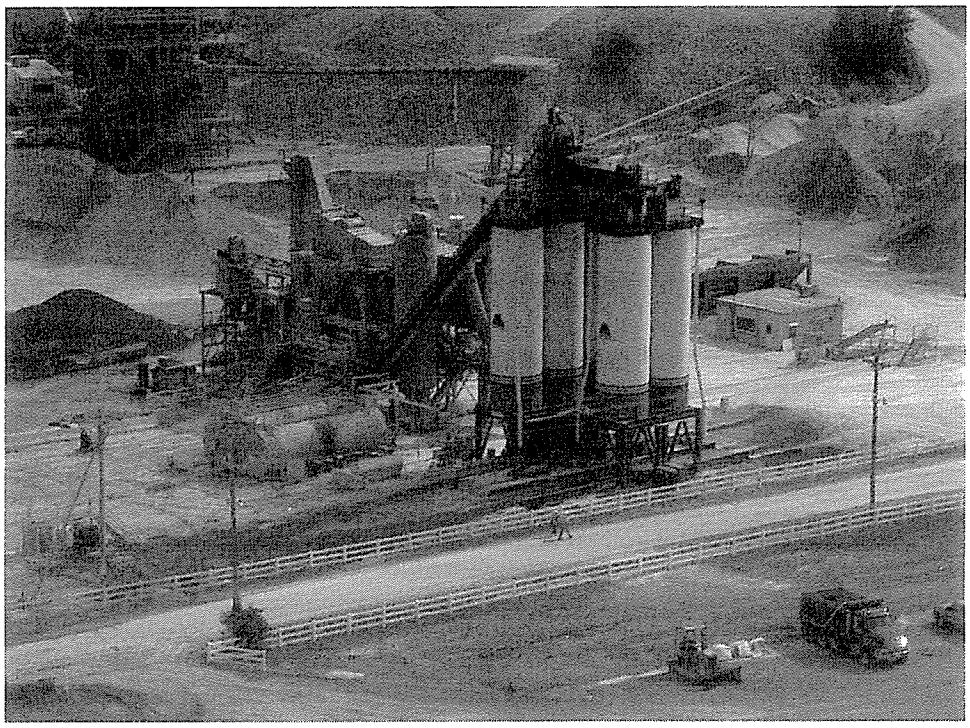


Figure 7.19 Modern Batch Plant Fitted with Recycle System, Baghouse, Filler Silo, and Mix Storage Silos

In today's world, sized aggregates are readily available. Mixes can be constructed with different percentages of sized aggregates at a ratio blended right from the cold-feed bins. This is one reason why drum-mixers came into being in the 1970s. It also eliminates some of the requirements for a batch tower. In fact, it is not uncommon today to find five or six different aggregates being blended at the cold bins to feed a batch plant. With an overwhelming majority (over 95 percent) of today's batch plants having only four hot bins, ratio blending at the cold feed affects the hot-bin gradations dramatically in a batch-plant operation.

Therefore, in modern batch-plant production, the blend formula is usually first established for the cold-feed bins to arrive at the target job-mix formula just like a drum-mix plant, and then the hot bins are sampled for their individual gradations and a new blend recipe/ratio is constructed with these hot-bin gradations to re-arrive at the job-mix recipe. The ratios of the individual hot bins become the recipe that is used to construct the final mix with the batching plant.

If the HMA producer has good quality control on the cold-feed bins, mixes can be proportioned completely with the cold feeds, eliminating the need to screen the material in the batch plant. This is often referred to as "screenless mixing." Screenless mixing is done by opening a bypass chute, allowing the heated aggregate stream to drop immediately into the No. 1 bin without passing over the screen deck. This requires drawing aggregate from only one hot bin when weighing an individual batch, which speeds up production. Bypassing the screen deck also eliminates wear and tear on the screens and the screening mechanisms, which are both expensive and time consuming to repair and maintain.

See Chapter 3, "Aggregates and Aggregate Additives," to review the mathematical procedure for establishing these blend ratios. This procedure is first used in the design process to establish the required percentages of each material for the job-mix formula; it is then used against the individual gradations of the cold-feed materials to reconfirm the blend ratios from the cold-feed bins. It is used again with the individual hot-bin gradations to establish the hot-bin percentages required to re-arrive at the target composite gradation.

Once these individual hot-bin percentages are determined, establishing the actual weights of each hot bin is relatively simple. The percentage for that hot bin is multiplied by the total quantity of aggregate desired in the batch and this becomes the target weight for that hot bin for that batch size.

Several things can negatively affect the gradations in the hot bins, which will therefore negatively affect the gradation in the final mix:

- A blocked or plugged cold-feed bin, or feeder, will reduce that material from the composite aggregate blend going to the dryer.
- If the cold-feed bins' blend ratios are set incorrectly, the gradations in the individual hot bins' will be negatively affected (too many coarse fractions in the bin or too many fine fractions in the bin).
- Too much material of any given size may exceed the screening and separating capability of the screens on the tower, causing that

material to also flow into the next larger bin. This is called carryover. (Gradations become finer in the final mix.)

- Exceeding the separating capability of the tower screens in total tons per hour (tph) will also cause this carryover effect. (Again, gradations become finer in the final mix.)
- Worn screens or screens with holes allow coarser materials to fall into the wrong hot bins. (Gradations become coarser in the final mix.)
- Screens that are caked or blinded over with clay, wet materials, or foreign materials reduce screening efficiency and potentially cause carryover. (Again, gradations become finer in the final mix.)
- Holes in the chutes onto which the screens discharge can cause larger or finer materials to go into the wrong bins. (Gradations can change either direction.)
- Holes in the bin walls that separate the hot bins can cause material to pass from one hot bin to the other. (Gradations can change in either direction.)

Some of these items are operational and some are maintenance related.

On the operational side, personnel should be aware of the separating capability of their screens. Figure 7.20 shows a typical flow configuration over a standard 3½-deck screen, illustrating how the material flows through the screens. Most screens are configured this way. The "½ deck" refers to the short screen positioned right below the top deck, or top screen. Sometimes this short screen is not half the length of the other screens, as illustrated. With 3½-deck screens, the ½ deck usually carries the same screen cloth as the beginning section of the very top deck, or screen. This provides the equivalent of a full-length screen-to-screen material for the No. 3 hot bin. (Hot bins are always numbered from the elevator out, or in the same direction that the material flows across the screen. Therefore, the No. 1 hot bin has the finest material in it, and the No. 4 hot

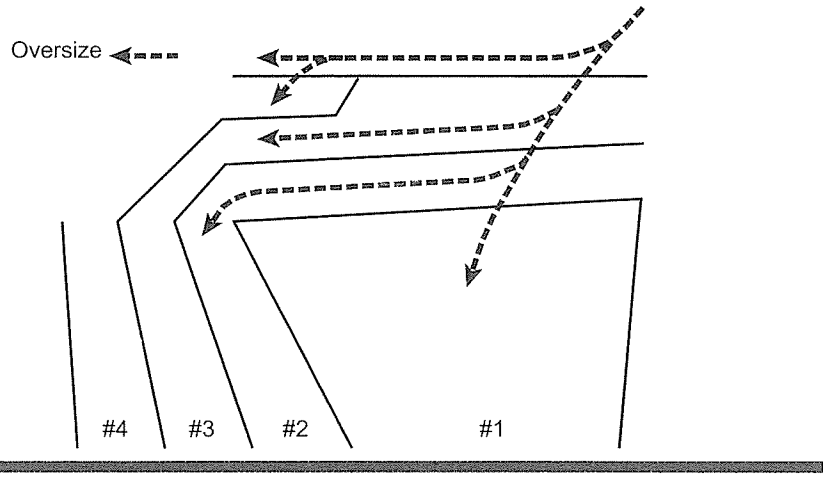


Figure 7.20 Material Flow Across a Typical Screen Deck

bin has the coarsest material. Remember that over 95 percent of the batch plants in North America have four hot bins. Some have three; some have five, but most have four.) The screen cloth for sizing material into the No. 4 bin typically has large openings. The No. 4 bin also typically receives a smaller quantity of material relative to the total feed, so it doesn't need a full deck for screening efficiency. Figure 7.2 and Figure 7.20 are artists' renditions of the flow of material across the screening unit and into the batch plant.

Different screen sizes can be used to fractionalize the material into different sizes, but most modern batch screens are set one time, representing the common mixes being made and are not often, if ever, changed. Typical screen sizes for plants at the time of this writing are No. 4 to No. 6 over the No. 1 bin (to create a sand-by-dust-type material), 7/16 inch over the No. 2 bin (to create a 3/8-inch-by-No. 4 or No. 6-type material), 5/8 inch over the No. 3 bin (to create a nominal 1/2 inch or 9/16-inch-by-7/16-inch-type material), and 1–1½ inches over the No. 4 bin (to create a nominal +¾ inch material—that being retained on the 5/8-inch deck, but passing the 1–1½ inch screen). Materials larger than 1–1½ inches are rejected out of the tower into the oversize chute.

Charts are available from batch-plant manufacturers that provide guidance on maximum screen capacity (before the material overflows) based on a given size opening. Slotted or z-wire screens can be used to improve throughput capability over standard square-screen types.

Sampling determines the gradations of the individual hot bins. Below the hot-bin gates on a batch plant is a door where a sampling device can be slid below the bin to catch the flow of material as it discharges from the gate. Figure 7.21 shows such a device.

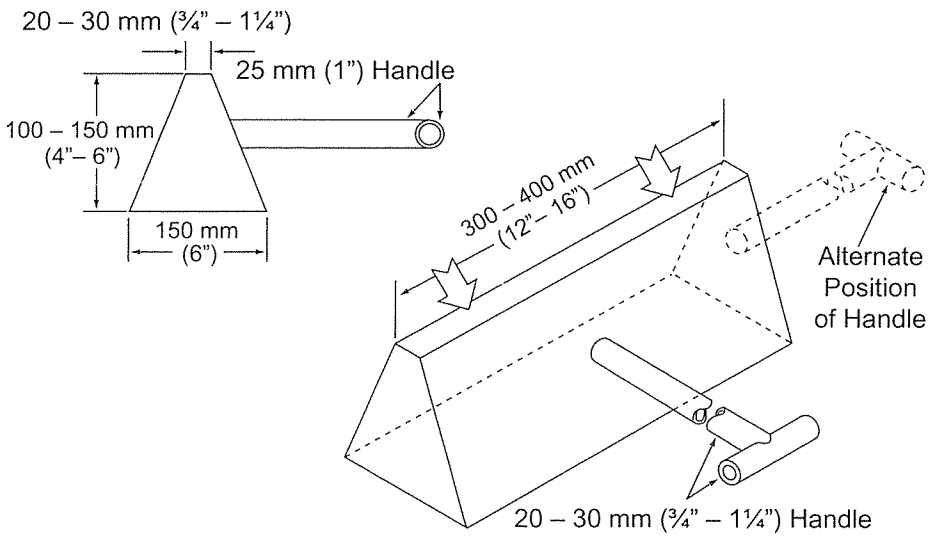


Figure 7.21 Hot-Bin Sampling Device

When sampling hot bins, it is important to take a full cross section of the flow from the gate because most hot-bin walls are sloped and materials can segregate slightly. Also, the flow of material across the screen lends itself to segregation. This does not cause any problems with the final mix because the materials are being mixed in the pugmill, but it is important to obtain an accurate gradation of each hot bin. Therefore, it is important to sample each bin across the full width.

After the sampling is done and the gradations from each hot bin are established, the ratio of each hot bin can be calculated to arrive at the job-mix formula.

Since material from each hot bin is weighed, the percentages of the hot bins need to be converted into weight values. Most batch computers make this job easy by allowing you to enter the formula in weights that total 1000 pounds (453 kilograms). For example, if hot bin No. 2 is 33.5 percent of the total mix, then the weight target in the batch formula will be 335 pounds (152 kilograms). The operator then selects the batch size—with the batch computer doing the calculations for the operator—to establish the weights for the entire batch. This procedure also allows the operator to change the batch sizes without changing the formula. It is common for batch plants to change batch size. This usually is close to the size typically run. Most computers make this adjustment. For example, if a plant is a 4-ton (3.6 metric tonnes) plant, but the desired load is 14 tons (12.7 metric tonnes), the batch computer will produce four drops at 3.5 tons (3.2 metric tonnes) to arrive at the target quantity.

The gradations in the final mix will only be accurate if the operator or the computer batches the proper quantity of material from each hot bin. The accuracy of the scale is critically important. Frequent calibration checks are advisable.

Each activity is required for gradation accuracy in the final mix produced with a batch plant. More aggregates in the job-mix formula and tighter mix tolerances require stringent management of the various factors that can affect final gradations.

7.3.2 Asphalt Content Control with a Batch Plant

Asphalt content control in a batch is simpler than gradation control. Asphalt is simply weighed to the target weight in each batch in a separate container called the asphalt weigh bucket. A fill valve opens, the computer watches the weight accumulate in the asphalt scale, and the computer stops the scale just before the target weight is reached, allowing some material to still drain from the valve and pipe until the target weight is reached. This early shutoff function is referred to as a free-fall compensation value, and is automatically adjusted by the computer so the exact target is reached with each batch. This same free-fall compensation feature is also used on

the aggregate scale for each hot-bin pull. The exact batch weights required would be difficult to achieve without this feature.

Once the target weight is achieved, the controls hold back the asphalt release until the dry-mix cycle in the pugmill is finished. Then a drain valve is opened (or a pump is engaged) to allow the asphalt to fall or to be pumped into the mixer. The drain mechanism remains open or on until the scale returns to zero.

The entire process of weighing and discharge is typically computer controlled.

7.3.3 Mixing/Coating with a Batch Plant

Aggregates and asphalt are mixed in a pugmill in a batch plant. The pugmill is a twin-shaft design, with adjustable hardened steel tips and hardened steel liners. Tips can be configured for different styles of mixing. Preferences vary, but all styles seem to be effective. The manufacturer's manual provides suggestions for setting the tip configuration. The pattern shown in Figure 7.22 is referred to as an around-the-box pattern. Tips adjust outward so that the clearance between the tip and the liner is not excessive. The tips are usually adjusted to slightly less than the size of the

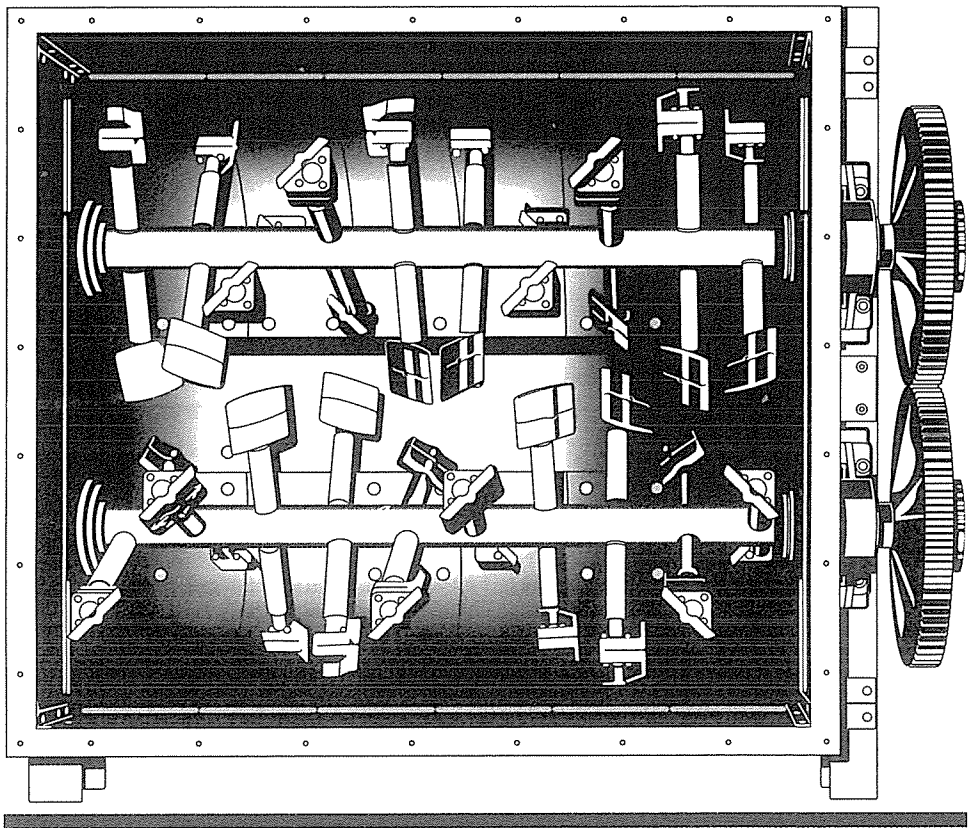


Figure 7.22 Twin-Shaft Pugmill Mixer Showing the Key Components to the Mixer

largest aggregate mixed and are then restored to target tolerance when the gap between the tip and the liner becomes excessive.

Missing tips, excessive clearance between the tips and the liner, and improper pattern to the tips all negatively affect the mixing efficiency of the pugmill.

Overfilling or underfilling the pugmill also causes inadequate mixing. When the pugmill is overfilled, the asphalt binder and the aggregates that were drawn last do not have a chance to be pulled down into the rest of the materials. Both asphalt content and gradation will be negatively affected and inconsistent throughout the batch. Significantly underfilling a mixer may also result in less than desirable results and cause segregation in large-rock mixes. A reasonable batch size should be respected.

Batch mixing time should be long enough to produce a homogenous mixture of evenly distributed and uniformly coated particles. Mixing time can be established for a particular mix by the procedure noted in AASHTO T 195 (ASTM D 2489). This method bases the degree of mixing on the percentage of coarse particles that are completely coated with asphalt. The method is correlated with mixing time.

The test involves separating coarse aggregate particles from the mix on a selected sieve size and examining 200 to 300 particles under a strong light. Any speck of uncoated aggregate particle visible to the naked eye is classified as uncoated. Usually particles coated at 90 and 95 percent are minimums for base and surface mixes, respectively. The least time needed to achieve these minimum coating levels is the most desirable mixing time. Most mixes will undergo additional mixing and coating as they are handled through slat conveyors, storage silos, and paving equipment. The common total batch mixing time is 25–35 seconds.

7.3.4 Recycled Mix Production with a Batch Plant

With a batch-type plant, two basic systems for recycled mix production are popular: the weigh-hopper method and the bucket-elevator method. With both systems, virgin aggregate is super-heated in the dryer and then the RAP is introduced later in the plant process. The super-heated virgin aggregates heat and dry the wet ambient RAP. The steam and dust released from the RAP has to be ducted to the exhaust system and the dust collection equipment. Because the RAP is not introduced into the aggregate dryer (as it is in a drum-mixer plant), additional ducting has to be added to the plant to accomplish this goal.

With the weigh-hopper method, RAP is either introduced into the aggregate-weigh hopper in the batch tower as an additional ingredient, or an additional weigh hopper is added specifically for the RAP. The control system weighs the RAP. One must adjust the binder content in the final mix recipe for the usable binder in the RAP (see Chapter 4, "Hot Mix Asphalt (HMA) Design").

With the bucket-elevator method, RAP is introduced with the virgin aggregate into the bucket elevator. The steam release starts in the bucket elevator, which results in a continuous steam release that is less immediate and less rapid than in the weigh-hopper method. This makes the steam ducting easier to design, install, and manage—explaining why this type of system is popular. With this method, however, the RAP is fractionated and separated into the different hot bins of the batch tower. This means that the asphalt content and the gradation of each hot bin must be analyzed so the proper weights for each hot bin and the new liquid binder can be properly determined. It is common to find belt scales fitted to both the virgin-aggregate conveyor feeding the dryer and the RAP conveyor feeding the bucket elevator. With these scales, the ratio of RAP to virgin aggregate can be maintained.

More information about producing recycled mixes in a batch-type plant is available from the National Asphalt Pavement Association in Information Series 123, "Processing and Handling Reclaimed Asphalt Pavements (RAP)."

7.4 The Drum-Mix Production Process

7.4.1 The Development of the Drum-Mix Process

Drum-mixers started appearing during the early 1970s. The first drum-mixers used elongated drying drums not only to dry and heat the aggregate but also to mix the liquid asphalt with the heated aggregate to produce the final hot mix all in the same drying cylinder.

These first drums were parallel-flow dryers, meaning that the aggregate and the process gas flowed in the same direction. The wet, ambient aggregates were introduced to the end of the dryer where the burner was located, and the aggregate was dried as it moved away from the burner. In this fashion, the liquid-asphalt binder could be introduced into the exit end/cooler part of the drum.

This process required less capital expenditure for initial equipment, greater mobility for relocation to remote projects, less installation time, less maintenance, and higher RAP capability.

This process could not have been developed without the following:

- The availability of uniformly produced, consistently sized aggregate
- The development of large, inexpensive variable-speed motors that can be fitted to the cold-feed bins to easily proportion aggregates to the desired final mix gradations
- The development of belt scales and flow meters for measuring the flow of materials continuously
- The development of finished mix storage silos to elevate the material so it can be dispensed into trucks

Each equipment development occurred independently and it became obvious that the need for sizing and fractioning material in a batch plant was no longer necessary.

Drum-mix plants were lower to the ground, easier to move, and less expensive to buy and maintain than batch facilities. As a result, they became the process of choice for hot mix asphalt producers starting in the late 1970s, and today over 80 percent of the mix produced in the United States is with drum-mix plants.

The drum-mixer has evolved since its early parallel-flow configuration days (Figure 7.23), to now include several slightly different styles of continuous processing:

- Counterflow drum-mixers (see Figure 7.24)
- Dryers with a separate continuous mixer—mixers can be either mixing drums or continuous pugmills (see Figure 7.25)
- Dryers with continuous mixers fitted together like the double-barrel plant—sometimes called a unitized dryer/mixer (see Figure 7.26)

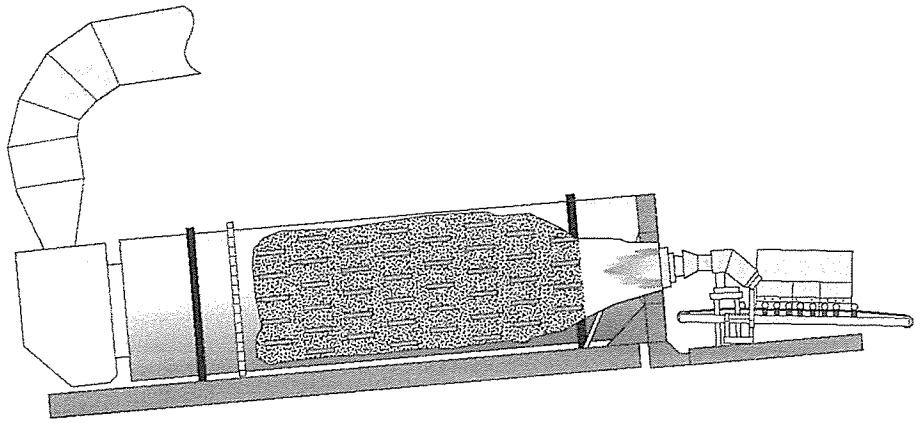


Figure 7.23 Parallel-Flow Drum-Mixer Flow

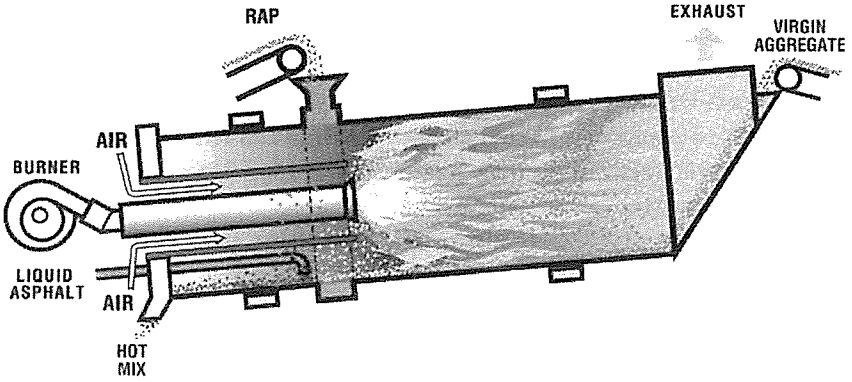


Figure 7.24 Counterflow Drum-Mixer Flow

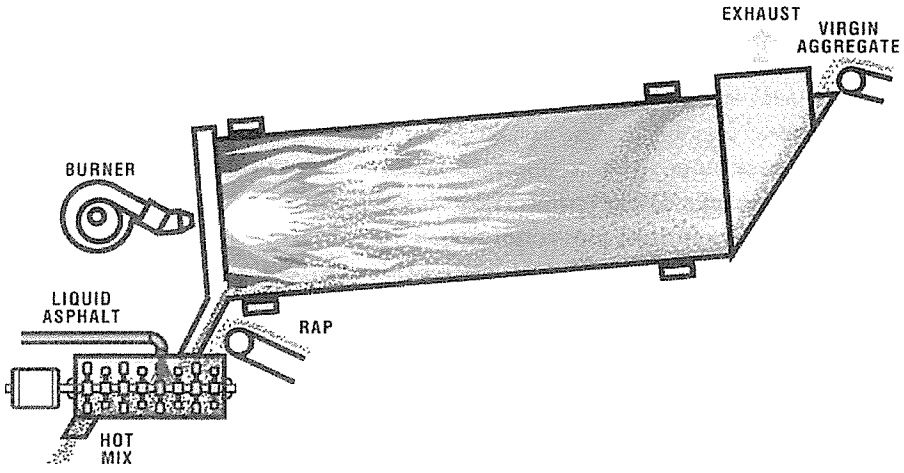


Figure 7.25 Separate Dryer and Mixer Flow